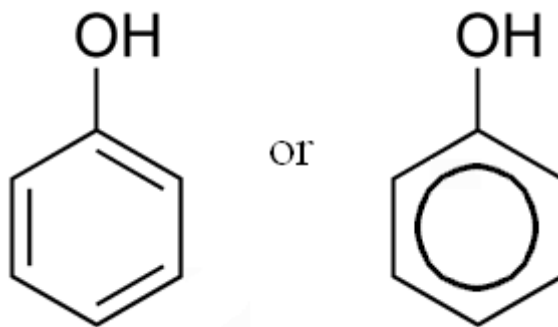


Phenols, Thiols, and Ethers

January 2026

Phenol

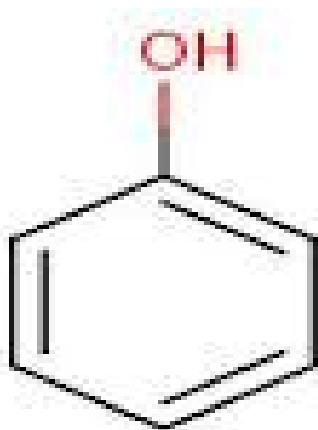


Phenols

In phenols, the hydroxyl group (OH) replaces a hydrogen atom in an aromatic ring.

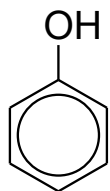
Many phenols are important as antiseptics and disinfectants.

The general formula Ar-OH.

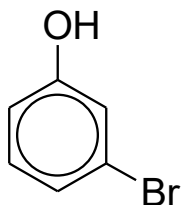


Nomenclature.

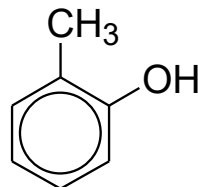
Phenols are usually named as substituted phenols. The methylphenols are given the special name, cresols. Some other phenols are named as hydroxy compounds.



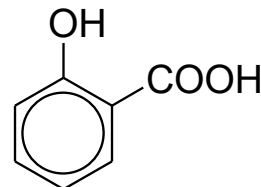
phenol



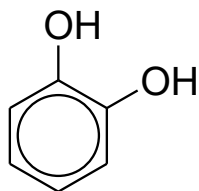
m-bromophenol



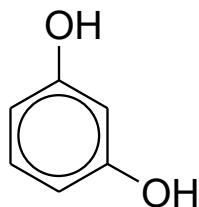
o-cresol



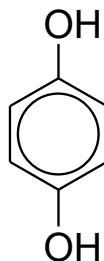
salicylic acid



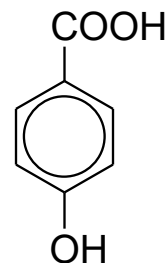
catechol



resorcinol



hydroquinone



p-hydroxybenzoic acid

Properties of Phenol

- At room temperature, phenol is a transparent, colorless, crystalline mass, a white powder.
- Phenol is soluble in alcohol, glycerol, petroleum and phenol is poisonous, corrosive, and flammable, affects the central nervous system and the liver and kidneys.

Ingestion of phenol

Symptoms may include burning pain in mouth and throat, abdominal pain, nausea, vomiting, headache, muscular weakness, increase in heart rate, irregular breathing, coma, and possibly death.

1-Cresols

Chemical structure of a cresol molecule has a methyl group substituted onto the benzene ring of a phenol molecule.

There are three forms of cresols that are only slightly different in their chemical structure and physical properties: ***ortho-cresol***, ***meta-cresol***, and ***para-cresol***.

Cresol is a better antiseptic than phenol and is less toxic. Commonly used as disinfectant.

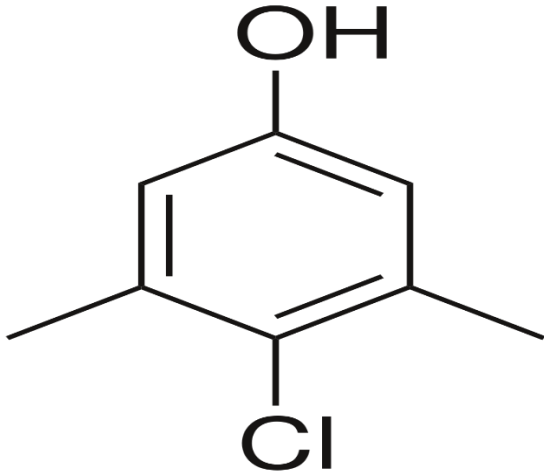
2- Resorcinol

Is M-dihydroxy benzene, it is also antiseptic, a resorcinol derivative
4-hexyl resorcinol is a much better antiseptic and germicide than resorcinol.

Chloroxylonol:

Chemical name: *para*-chloro-*meta*-xylonol

Dettol



Chloroxylenol, also known as *para-chloro-meta-xylene*ol

Is an antiseptic and disinfectant which is used for skin disinfection, and together with alcohol for cleaning surgical instruments.

It is also used within a number of household disinfectants and wound cleaners. It is thought to act by disrupting microbial cell walls and inactivating cellular enzymes, and is less effective than some other available agents. It is available as a liquid.

Antiseptics and disinfectants:

Both widely used to control infections.

They kill microorganisms such as bacteria, viruses, and fungi using chemicals called biocides.

Disinfectants are used to kill germs on nonliving surfaces. Antiseptics kill microorganisms on your skin.

3-Some oils and plants,

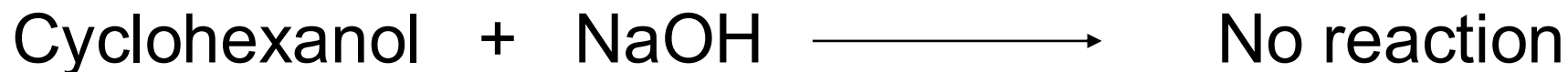
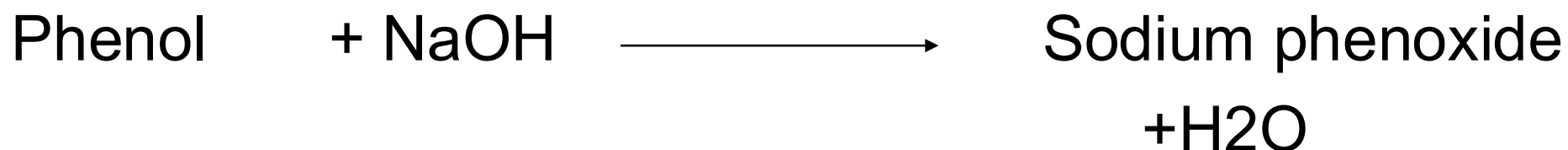
Which produce the odor and flavor of the plant, are derivatives of phenol, for example;

Eugenol	is found in cloves
Isoeugenol	is found in nutmeg
Thymol	is found in thyme

Reactions of phenols

1-Reaction with aqueous sodium hydroxide solution

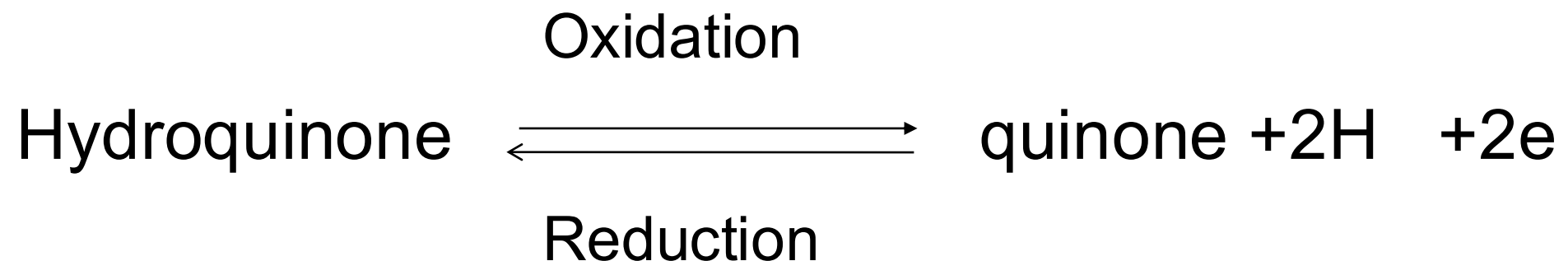
To form sodium phenoxide, alcohols such as ethanol and cyclohexanol do not react with basic solutions, and this is due to phenols are weak acids but alcohols are not.



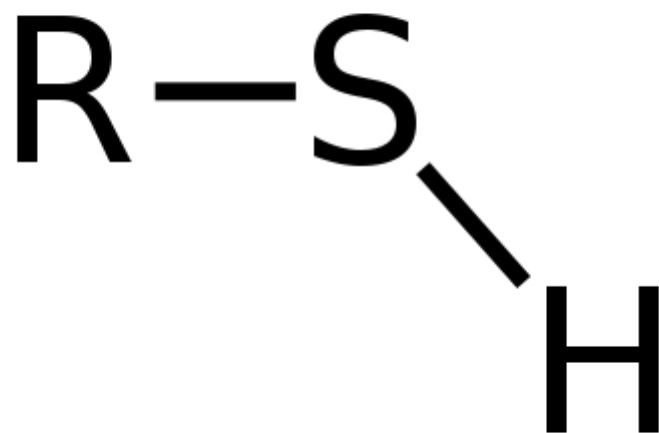
2-Oxidation-reduction reaction

Phenols such as hydroquinone (1, 4-dihydroxy benzene) undergo an important oxidation-reduction reaction.

Hydroquinone is easily oxidized to quinone. This reaction is reversible, because quinone is easily reduced to hydroquinone.



Thiols



Thiols R-SH

Thiols are compounds sulfhydryl (**-SH**) group replaces a hydrogen atom in an alkane, for the names, thiol is added to an alkane.

CH ₃ -SH	Methane thiol	HS-CH ₂ -CH ₂ -SH
		Ethane dithiol

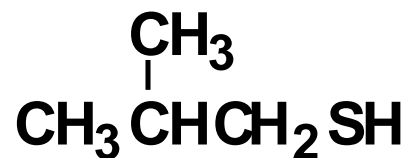
Many thiols are found in nature, and they all have a disagreeable odor.

Common name

- Common names for simple thiols are derived by naming the alkyl group bonded to -SH and adding the word "mercaptan"



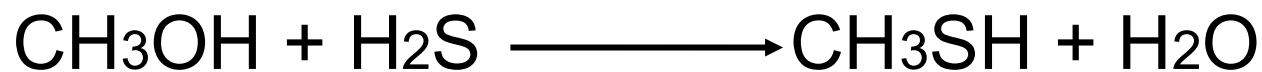
Ethanethiol
(Ethyl mercaptan)



2-Methyl-1-propanethiol
(Isobutyl mercaptan)

Preparation of Thiol

- In industry, methanethiol is prepared by the reaction of hydrogen sulfide with methanol. This method is used for the industrial synthesis of methanethiol:



Examples;

$\text{CH}_3\text{CH}_2\text{CH}_2\text{-SH}$ 1-Propanethiol in an onion

$\text{CH}_2=\text{CH-CH}_2\text{SH}$ 2-Propenethiol in garlic

$\text{CH}_3\text{-SH}$ Methane thiol in oysters and
cheese

Note;

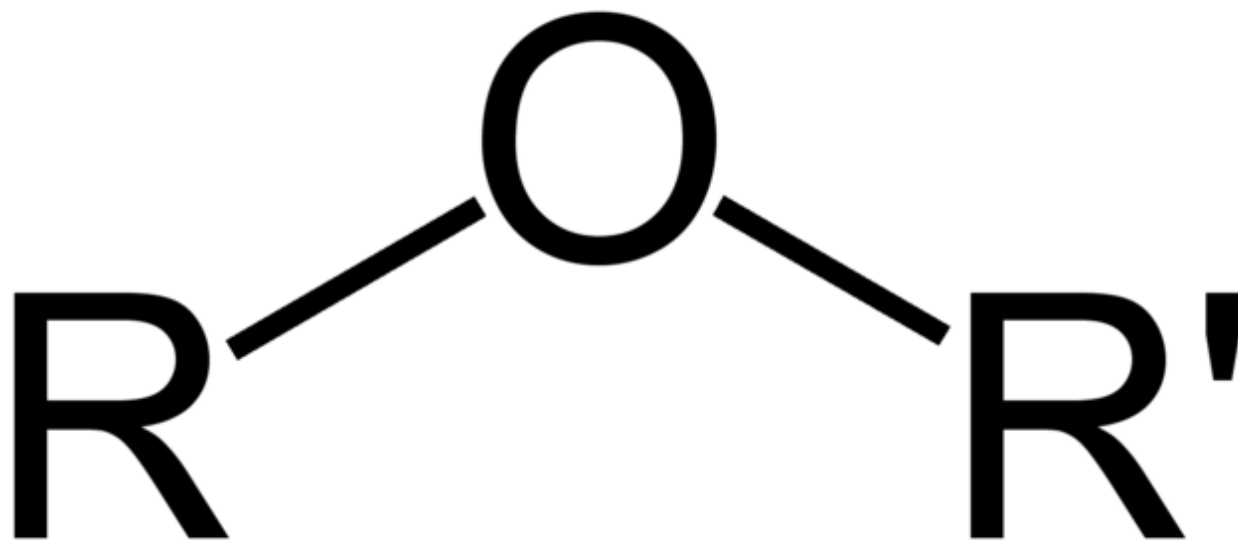
To help as detect natural gas (Methane for heating and cooking) leaks, a small amount of ethane thiol is added to the gas supply.

Physical properties;

Unlike alcohols, thiols do not exhibit hydrogen bonding. Therefore they have lower boiling points than the corresponding alcohols.

Also, they do not exhibit hydrogen bonding, thiols are less soluble in water than alcohols with the same number of carbon atoms.

Ethers



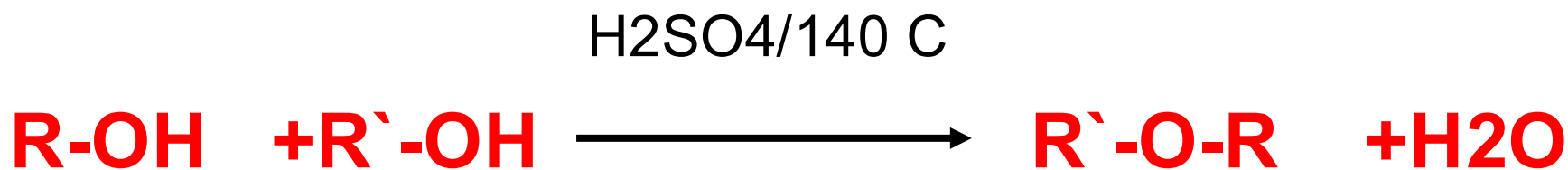
Ethers

- Ethers are characterized by the R-O-R functional group.
- They are generally non-reactive, but are extremely flammable.
- Diethyl ether was the first general antiseptic used in medicine. It has been replaced by **Penthrane** and **Enthane**, which are less flammable.

Ethers; $R-O-R'$ $Ar-O-R$ $Ar-O-Ar$

- **Ethers** are compounds that contain an oxygen atom bonded to two alkyl groups, two aryl groups, or one aryl and one alkyl group.

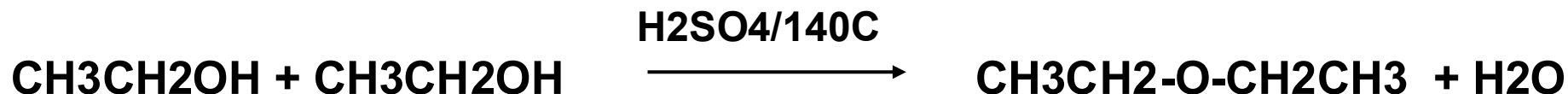
Preparation of ethers;



Diethyl ether

Diethyl or ethyl ether: a colorless, transparent, very volatile, highly flammable liquid with a characteristic odor; it was the first inhalational anesthetic used for surgical **anesthesia**, in United state and Kanada used but because easily flammable, avoided uses as anesthetic substance.

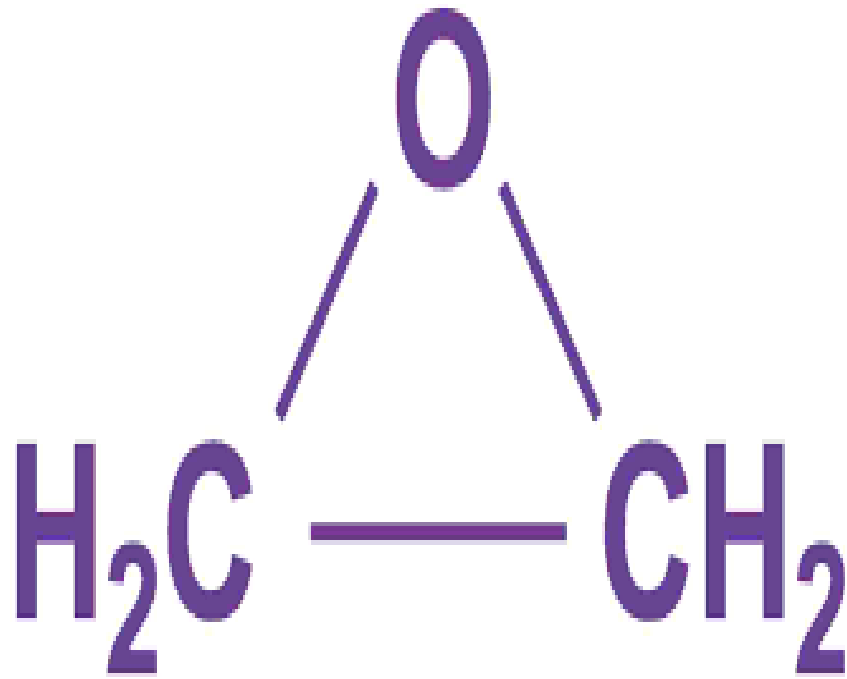
Preparation of Diethyl ether:-



Ethers are organic compounds very inactive, so that are not used to reaction, but used as solvents.

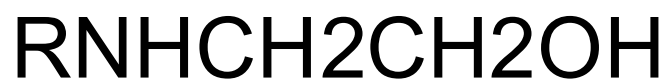
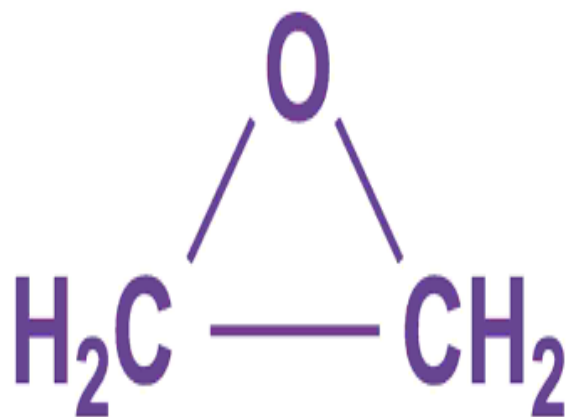
Some ether oxygen atom is placed in the ring called cycloethers Like tetra hydro furan (THF).

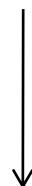
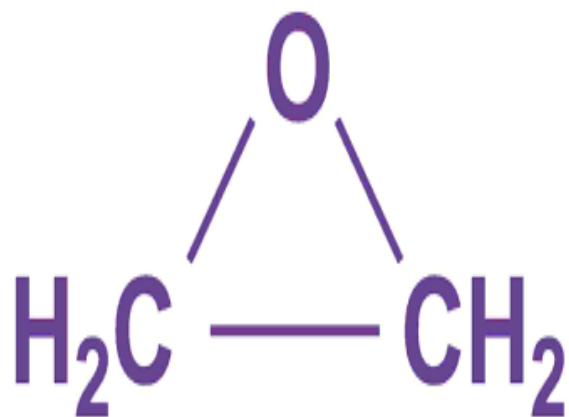
Ethylene oxide: a chemical sterilant



Ethylene oxide: a chemical sterilizing

Ethylene oxide is a colorless, flammable gas with a boiling point of 11°C . Ethylene oxide reacts with the amino $-\text{NH}_2$, and sulfhydryl $-\text{SH}$ groups present in biological materials. At sufficiently high concentrations, it reacts with enough molecules in cells to cause the death of microorganisms. This toxic property is the basis of ethylene oxides use as a fumigant in foodstuffs and textiles and its use in hospitals to sterilize surgical instruments.





Physical properties of ethers

Ether molecules cannot form hydrogen bonds with each other, resulting in relatively low boiling points compared to those of the analogous alcohols.

For Hydrogen bonding, the Hydrogen must be bonded with some strongly electronegative atom (F,O,N). But in ethers the bonding is R-O-R. All the hydrogens are attached to carbons and no H is attached with the electronegative oxygen. This prevents intermolecular Hydrogen bonding.